



### App to solve equations

PhotoMath is an app which uses the phone's camera to solve equations. Currently for iOS and Windows <http://dgit.in/AppPhotoMath>



### Hybrid brain sensors

DARPA and the University of Wisconsin have built hybrid brain sensors to map neural activity <http://dgit.in/BrainSensor>

# Digit 101 – Monitor interfaces

We demystify everything that a display needs as far as input/output is concerned. Let's look at all the different interface options available under the sun

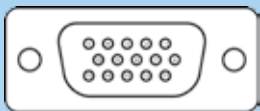
Siddhant Sharma

[readersletters@digit.in](mailto:readersletters@digit.in)

Ever wondered why there are so many different types of analog and digital connectivity options on a display? What do they look like and how do you differentiate between them on the basis of bandwidth? Or what's the maximum resolution a certain signal standard is capable of? Worry not, we're here to guide you through each of the popular signal standards and their respective connectors and also speak about what's in store for them in the future.

## VGA

Designed in 1987, Video Graphics Array (VGA) is a signal standard whose connector is also christened with the same name. It has become the



The DE-15 socket pin out

universal analog display interface. What makes VGA different from DVI and now slightly outdated is that it carries an analog signal. Although it supports a resolution of 2048 x 1536, when was the last time you saw a VGA monitor running at that res?

The all-too-familiar blue VGA connector is dubbed DE-15 because it has 15 pins. Don't take our word for it, go on and count 'em.

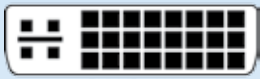
As it's been almost three decades since VGA came into being, industry top guns such as Intel, AMD, Dell, Lenovo, Samsung and LG have already signed on to phase out this de facto display interface in order to let HDMI and DisplayPort flourish (Read more about it at [dgit.in/VGA\\_Death](http://dgit.in/VGA_Death)).

## DVI

After VGA, DVI was the answer to improving display interface technology



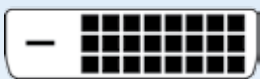
DVI-I (Single Link)



DVI-I (Dual Link)



DVI-D (Single Link)



DVI-D (Dual Link)



DVI-A

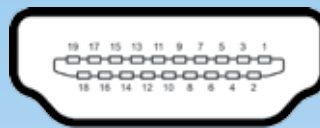
Different DVI connector types

in general, since it was not only digital, but analog as well – which meant that it featured support for analog connections and hence compatible with the older VGA interface. Being the best of both worlds combined in a decent 29-pin package, DVI is capable of carrying RGB data, clock and display data channels together at an amazing bitrate of 3.96 Gb/s in single-link mode. Now, what the heck is 'single link'? Well, DVI primarily supports multiple modes such as Digital only, Analog only and even Digital plus Analog, hence we have the DVI-D, DVI-A and DVI-I connector options, respectively. The digital only DVI-D, for instance, features single-link and dual-link varieties. The single link employs a lone 165Hz transmitter that supports a resolution of up to 1920 x 1080

@ 60 Hz, but the dual-link mode makes use of a second transmitter with six additional pins and bumps up the resolution to 2560 x 1600 @ 60Hz. A resolution of 3840 x 2400 @ 33Hz is also supported. These six pins at the centre of a dual-link DVI-I or DVI-D connector are absent in a single-link connector and DVI-A skimps on 12 pins at the centre in total and that's how you can identify a particular one.

## HDMI

In a collective bid to throw analog out of the window, companies such as Hitachi, Panasonic, Philips, Sony, Toshiba and others came together to



19-pin HDMI pin out in its full glory

create HDMI or High-Definition Multimedia Interface that revolutionised the display industry. It provided a compact, yet extremely functional interface for transferring uncompressed digital video data and compressed or uncompressed digital audio data between a HDMI-

compliant display controller and display device such as computer monitors and TVs. The key difference between HDMI, DVI and VGA is that HDMI completely lacks VGA compatibility as it doesn't have any analog contacts. In spite of this and many other differences, in order to provide backward compatibility between DVI-D and HDMI, most HDMI devices support DVI-D signalling too.

Various versions of HDMI have been deployed till date. Most current devices use versions 1.2, 1.3(a) or 1.4, while HDMI 2.0 is the latest version. HDMI 1.4, with its 19-pin connector, is capable of carrying a maximum resolution of 4096 x 2160 @ 30Hz. Pixel clock increments in different HDMI versions have allowed to tap into HDMI's potential, thus allowing for transmission of 3D 4K content at high frame rates through the new HDMI version 2.0.

## DisplayPort

The ultimate digital video transmission interface, DisplayPort, is basically the first ever interface to depend on packetised data transmission like



A typical DisplayPort connector

the one used by Ethernet. What this means is that it can allow for transmission of extremely high quality uncompressed content through fewer pins (20) as compared to other interfaces. DisplayPort version 1.3

now enables transmission at a whopping bandwidth of up to 32.4 Gb/s, thus making it possible to run multiple displays off a single DisplayPort connector. Although 4K has just started getting mainstream, this crazy amount of bandwidth provided by DisplayPort 1.3 allows for transmission of 4K 3D, 5K and even 8K (7680 x 4320) content. If the price of 4K displays hasn't put a dampener on your hopes of owning one, this surely will. After all, a beefy flagship today is the runt of the family tomorrow in the tech world. 